

ULK POLYTECHNIC INSTITUTE

Technical Report: Cryptography & Network Security

Module: ETTCS801

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1 Risk Assessment

A comprehensive security review of the polytechnic institute's infrastructure revealed critical vulnerabilities exposing central student records.

1.1 Assets, Vulnerabilities, and Consequences

1. **Central Student Records Server:** Susceptible to cleartext network eavesdropping and unauthorized guest access due to unencrypted transfer protocols.
2. **Staff User Credentials:** Weak password standards expose system endpoints to brute-force credential harvesting.
3. **Core Server Host:** Outdated software updates and missing firewall perimeter filters permit probing from external unknown networks.

1.2 Risk Evaluation and Mitigation Controls

Risk Event	Likelihood	Impact	Rank	Recommended Control
Data Interception	High	High	1	Deploy <code>iptables</code> filtering rules and enforce SFTP/HTTPS protocols.
Credential Exploitation	High	Medium	2	Mandate multi-factor authentication (MFA) and strong password rules.
Software Exploitation	Medium	High	3	Apply automated security updates and restrict edge inbound access.

Table 1: Risk Assessment Ranking and Control Measures

2 Encryption and Integrity Application

To ensure record confidentiality and detect data tampering during campus transfers, a Python security toolkit was implemented using symmetric Fernet encryption (AES-128 in CBC mode) and SHA-256 digest hashing.

2.1 Functional Overview

- **Key Isolation:** The encryption key is generated locally and stored in `secret.key`, which is ignored by Git.
- **Decryption Verification:** Decrypted payload content is cross-checked against original records to verify exact content restoration.
- **Integrity Verification:** Computes 256-bit hashes to detect unauthorized modifications.

2.2 Execution Test Evidence

During toolkit execution on `sample_student_record.txt`, the following verification outputs were observed:

```
1 --- Baseline SHA-256 Hash ---
2 [+] SHA-256 Digest: 326984
   c9132f0e92b2cd9e6d47313f3ab5af9b343bd76afbee5bf7f50a260720
3
4 --- Integrity & Tamper Detection Test ---
5 [*] Testing unmodified original file:
6 [+] Integrity Verification PASSED: File is intact and unchanged.
7
8 [*] Testing modified file 'tampered_student_record.txt':
9 [+] SHA-256 Digest:
   f8544deb360f6fe68b1ef671dbade034d3044e98908a639c55d3946d20257fb7
10 [!] Integrity Verification FAILED: File alteration or tampering detected!
```

3 Network Traffic Filtering

Perimeter isolation was enforced on the laboratory host using Linux `iptables`.

3.1 Applied Firewall Rules

```
1 # Block guest network access to port 443
2 iptables -A INPUT -s 192.168.20.0/24 -p tcp --dport 443 -j DROP
3
4 # Permit staff network access to port 443
5 iptables -A INPUT -s 192.168.10.0/24 -p tcp --dport 443 -j ACCEPT
6
7 # Block remaining inbound access to port 443
8 iptables -A INPUT -p tcp --dport 443 -j DROP
```

3.2 Test Procedure and Results

Source Network	Destination Port	Expected Outcome	Actual Result
Staff Workstation (192.168.10.0/24)	443 / TCP	Allow Connection	Connection Succeeded
Guest Device (192.168.20.0/24)	443 / TCP	Drop Connection	Connection Timed Out
External Machine (172.16.0.0/16)	443 / TCP	Drop Connection	Connection Timed Out

Table 2: Firewall Rule Test Results

4 Conclusion and Submission Details

The implemented toolkit successfully protects student record files at rest and in transit via strong symmetric encryption, verifies data authenticity using cryptographic hashes, and blocks unauthorized network traffic.

GitHub Repository URL: <https://github.com/Dodos-N/cryptography-network-security-exam>